SYSTEM ARCHITECTURE

DECIXN — Market Intelligence Platform

Advanced Financial Ecosystem · Agentic Workflows · Institutional Analytics

CLIENT LAYER

Interactive Dashboard

- React, Vite, TS, TailwindCSS, ApexCharts, Recharts
- Custom Portfolio Upload, Insights, Watchlists, Stock/MF Details, AI Chat, Alerts, News

Foxy v1 — Advisory Chatbot

Financial co-pilot · Portfolio-aware reasoning · Session persistence · Streaming Intermediate Steps

AI CORE — FOXY V1

Google Gemini 2.5 Flash

Multi-tool reasoning core · Logic synthesis · Narrative generation

Tool-Augmented Library

Analyze Ticker · Fetch Portfolio · Fetch Watchlist · Get Positions · Fetch Relevant News · Display Charts

HTTP / WebSocket

Gemini API calls

API & ORCHESTRATION LAYER

FastAPI Core Routing

REST API Endpoints

News Scraper Engine

ET Market Feed · Catalyst Processing · 10m Ingestion Cycle

Real-time Alert Engine

Price/RSI/Signal triggers · Push Notifications · Real-time updates

orchestrates ↕ enriches

AGENTIC INTELLIGENCE LAYER

LangGraph Workflow Engine

Uses AI Core Foxy + Tools for full-loop ticker analysis

Signal Detection (Deterministic Tools) → Context Enrichment (News / Portfolio Tool) → Verdict Node (Gemini 2.5 Logic)

CORE ANALYTICS & CATALYSTS

Technical Matrix

RSI, MACD, BB, VWAP, EMA (10-200), Pivot Points

Signal Discovery

Breakouts, Mean Reversion, Momentum, Rel. Strength

Overall Market Analysis

Index health · Sector movers · Heatmaps

Opportunity Radar

High-conviction scans · Dividend trackers

Decision Engine

Weighted verdicts · Risk-reward scaling

Catalyst Engine

Sentiment scoring · Multi-source correlation

Alpha Signal Fusion

LIVE NSE India PIT Disclosures · Corporate Filings

Statistical Range Projector

ATR Volatility Core · Multi-Factor Range Projection

Multi-Factor Skew Engine

RSI & Momentum Skew · Mean Reversion Pricing Bias

persists ↕ fetches live market data

DATA & PERSISTENCE LAYER

Supabase / PostgreSQL

Centralized storage for Portfolios, Alerts, Chat History, and Signal Caching.

yfinance API

Primary feed for Stocks and Mutual Fund market data (NSE/BSE).

AMFI India

Mutual Fund listing registry for ISIN mapping and Discovery Search.

ET Market

Latest News Feed ingestion for live Catalyst and Sentiment analysis.

NSE India

Direct Pit Disclosures & Corporate Announcements API scraper (Session-Aware).

REACT + VITE · TYPESCRIPT · TAILWINDCSS · FASTAPI · LANGGRAPH · GEMINI 2.5 FLASH · SUPABASE (POSTGRESQL) · YFINANCE · RECHARTS + APEX · NSE INDIA · ET MARKET NEWS · AMFI INDIA · PYTHON 3.11

AGENTIC CHAT WORKFLOW

Foxy v1 — Chat Request Lifecycle

ChatEngine · Tool Router · LangGraph Agent · Gemini 2.5 Flash · Supabase

FILE chat_service.py + agent_engine.py
MODEL gemini-2.5-flash
FRAMEWORK LangGraph + FastAPI

User / Client · Gemini LLM · Tool Execution · LangGraph Agent · API / Orchestration · Data / Persistence

① INPUT & CONTEXT ENRICHMENT

1. User Message Input

Raw query containing @mentions:

"Is my @portfolio safe if @stock:NHPC drops 5%?"

"Can you tell me if there are good buys in @watchlist:IT_AND_BANKS?"

@portfolio · @watchlist · @stock

yield { status: "Thinking…" }

② FIRST GEMINI CALL — INTENT & TOOL SELECTION

Gemini 2.5 Flash — Turn 1

Receives **SYSTEM_INSTRUCTION** (role, topic enforcement, output schema) + enriched prompt + **TOOLS declaration**.
Decision: "Is this query a market data request or a general conversation?"

Tool declarations

analyze_ticker · get_market_overview · analyze_full_portfolio · get_user_position · fetch_catalyst_news · fetch_watchlist

response.candidates[0].content.parts

③ RESPONSE BRANCH DECISION

function_call present?

NO — direct

Direct Text Response

Gemini returns a plain JSON narrative (off-topic, general advice, clarifications).

ACTION: Drafting a friendly response for off-topic inquiries.

type: general · type: off_topic

Persist + Yield Response

_save_assistant_message() → Supabase persistence. JSON yielded to WebSocket.

YES — tools

Status Emit → Client

Yields context-aware streaming status before dispatching tools.

TRANSPARENCY: Informing user what the agent is checking.

"Analyzing stock…" · "Checking portfolio…"

_execute_tool() — Parallel Dispatch

Iterates all tool requests.

EFFICIENCY: Multiple markets tools running in parallel threads.

④ TOOL EXECUTION DETAIL

analyze_ticker
Calls **analyze_single_ticker()** for indicators.
Technicals: RSI/MACD detection and trend validation.

get_market_overview
Returns NIFTY / SENSEX levels and Index health.
Market Health: Checking broad market sentiment and breadth.

analyze_full_portfolio
Regex-parses all holdings from portfolio context string. Routes to **run_portfolio_analysis()** (stocks) or **run_mf_portfolio_analysis()** (mutual funds). Returns P&L, win rate, weights.

get_user_position
Regex-scan of portfolio context for ticker match.
Personalization: Customizing advice based on your current buy price.

fetch_catalyst_news
Scrapes catalysts from ET Markets with sentiment.

AlphaSignalService
Scrapes **Live NSE India** for Insider Trades & Filings.
Direct Alpha: Capturing M&A, Dividends, and Promoter buying.

⑤ LANGGRAPH AGENTIC ENGINE — RUN_AGENT_INTELLIGENCE()

Triggered only by analyze_ticker tool

StateGraph — AgentState

Typed state: symbol, data, news, signals, context, portfolio, verdict, errors

detect_signals_node
Fetch data → Compute Indicators (RSI/MACD)

→

enrich_content_node
Parse News + **Live NSE Alpha Signals**

→

generate_verdict_node
Gemini synthesis → JSON Verdict

STRONG BUY / BUY / HOLD / SELL · confidence 0-100 · entry · exit · stop_loss · portfolio_action

agent_intelligence merged into tool_result

⑥ SECOND GEMINI CALL — NARRATIVE SYNTHESIS

Gemini 2.5 Flash — Turn 2 (Final)

Receives full conversation contents + **all tool_results**.

SYNTHESIS: Weaving technical data and news into a professional research note.

Output Schema enforced

stock_analysis · portfolio_analysis · market_overview

metadata includes:
tickers, charts, sources, summary, insight, sentiment, reasoning_steps

yield { status: "Generating final insights…" }

⑦ OUTPUT PARSING, PERSISTENCE & DELIVERY

_parse_json()

Strips markdown fences. Finds boundaries. Falls back if parse fails.

Supabase Save

Writes narrative + metadata to session history.

Yield Final

JSON stream to React.
INSIGHT: Actionable.

INTELLIGENCE ENGINE

News Catalyst Engine

Multi-factor event extraction & sentiment pipeline.

① EVENT INGESTION & SCRAPE

Data Ingress pipeline

Stateless polling nodes trigger every 10m.
Checks ET Markets latest news data, processes maximum of 5 new ones each time.

Raw HTML / JSON Stream

② EXTRACTION & ENTITY MAPPING

Regex-Driven Categorization

Stripping HTML tags and normalizing timestamps. Detecting **Stock Mentions** via Ticker Regex cross-ref.

CATALYST CLASSES

- EARNINGS
- DIVIDEND
- IND_DATA
- RATINGS_CHG
- M_AND_A
- GEN_MARKET

Clean Event Objects

③ SENTIMENT SCORING

Gemini Flash 2.5 — Contextual Scorer

Evaluates news impact magnitude (0-100) and directional bias. Unlike simple keyword matching, the LLM understands market nuance.

LOGIC: "Will this news trigger immediate retail buying or institutional sell-offs?"

BULLISH/POSITIVE	NEUTRAL	BEARISH/NEGATIVE

Score Array [Sentiment, Magnitude]

④ PERSISTENCE & CACHE

Supabase Persistence

Stores findings in the **catalysts** table. Cached for 15-30m at the Edge to prevent redundant LLM scoring costs.

```
UPSERT INTO news_cache (headline_hash, sentiment, tickers)
```

WebSocket Dispatch

Downstream Consumption

Findings are actively injected into **Foxy v1 Chat Logic** & also passed in the **News Page** & official **Stock Page** for immediate retail consumption.